How ERES Facilitates AI Development

CyberRAVE 72 Key Domains — AI Specialization, TETRA Curriculum & Common-Core BEE Framework

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1. Executive Summary

The ERES Institute for New Age Cybernetics provides five structural roles in AI development: (1) Validation Architecture via MIEVM; (2) Authentication Protocol via EAAP/PoR; (3) Value Accrual via Meritcoin/BERA; (4) Deployment Infrastructure via SCALULAR/H2C2H; and (5) Legal & Prior Art Containment via the Constitutional Gift Thesis. The CyberRAVE 72 Key Domains are the universal deployment map across which all five roles operate — every domain is simultaneously a SCALULAR node, a MIEVM validation theater, and a CyberRAVE resonance field.

This v2.0 document integrates three additional architectural layers absent from v1.0: the TETRA Curriculum (RT_Media delivery across all four SCALULAR pillars, Levels 101–401, 48 modules, 1,920 hours); the STORM_PARTY-(AI and Future Technologies) programming strand at the 301 Applied Integration level; and the Common-Core BEE stack (Bio-Ecologic Economy with THOW/HFVN/FDRV/GSSG operational instruments). Together these layers transform the document from a mapping reference into a full deployment curriculum with governance architecture.

2. The Five Structural Roles of ERES in AI Development

2.1 Validation Architecture — MIEVM

The Multi-Instrument Ensemble Validation Method requires convergent scoring across independent instruments before a GO threshold (≥8.5/10) is declared. The Erasure Completeness Ratio (ECR) is the convergence metric. The current ensemble — Claude, Grok, DeepSeek, ChatGPT — was culled from a larger set by the same constraint-based epistemology that produced the CyberRAVE 72. Single-instrument AI evaluation is constitutionally insufficient under ERES governance.

2.2 Authentication Protocol — EAAP / PoR

Proof-of-Resonance: PoR = A×R×P×F (Authenticity × Resonance × Purpose × Fidelity). Conjunctive — all four factors must satisfy. No compensatory scoring. The Runtime Containment Requirement and DSAP-PRE complete the protocol. Applicable to all agentic AI deployments across the 72 domains.

2.3 Value Accrual — Meritcoin / BERA

BERA indices (ARI, ERI, RHC, RCI) measure AI performance in resonance terms, not accuracy alone. Meritcoin operationalizes this on Proof-of-Resonance consensus: it is not mining, it is tuning. The BEE (Bio-Ecologic Economy) is the economic substrate; UBIMIA is the universal access layer. Students in the TETRA curriculum earn Meritcoin while learning — the curriculum is self-funding.

2.4 Deployment Infrastructure — SCALULAR / H2C2H

The four SCALULAR pillars (HEALTH/SSHP, LAW/SSLA, PROTECTION/SSPS, SKILLS_TRADE/SSST) define the institutional channels through which AI reaches citizens. All Tier 1 = SSSC (Solid-State Smart-City Citizen). H2C2H (Human-to-Computer-to-Human) is the AI development methodology: the ensemble is the epistemological guarantor, not the single model. RT_Media is the curriculum delivery channel at 301 level and above.

2.5 Legal & Prior Art Containment — Constitutional Gift Thesis

ERES functions as prior-art boundary and legal containment architecture for AI in the information commons. Three-instrument legal stack: Deterministic Reference (Daubert-load-bearing), MIEVM-LLM (governance commentary), BERA (governance tuning). Petition for Rulemaking pathway: 16 CFR §1.31 (FTC) / 12 CFR §1075 (CFPB). The corpus of 1,060+ PDFs and 456+ ResearchGate deposits under ORCID 0000-0001-9946-3221 constitutes prior art at scale.

3. TETRA Curriculum — RT_Media: Health · Law · Protection · Skills/Trade + Certification

The TETRA Curriculum is the four-pillar certification architecture of the Solid-State Smart-City, delivered through RT_Media across Levels 101–401. Total: 48 modules, 1,920 hours. Each module is deliverable through PlayNAC, measurable through BERA, credentialed through EarnedPath (EP=CPM×WBS+PERT), and recorded on GraceChain. The curriculum is self-funding through UBIMIA: students earn Meritcoin (at level-appropriate yield multipliers) while learning. The MIT/SPRU/BAIDU palindromic bridge provides the institutional frame: a student entering from the West and a student entering from the East traverse structurally identical content and emerge with the same SSSC credential, because C=R×P/M is invariant to direction of approach.

3.1 STORM_PARTY-(AI and Future Technologies) — Level 301 Programming Strand

The 301 Applied Integration level is the level where theory meets deployment. STORM_PARTY-(AI and Future Technologies) is the named programming strand that runs through all four SCALULAR pillars at Level 301, carrying AI and emerging-technology content through the Common-Core BEE alignment. Storm Party at 301 is not emergency management alone — it is the architecture's recognition that AI-and-future-technology integration is inherently a storm-condition operation: high stakes, real consequences, requiring multi-instrument validation, collective weathering, and crisis-grade merit recognition for those who do it well.

The 301 level carries a 3.0× Meritcoin yield multiplier — the highest of the taught levels, below only the 401 Advanced Practitioner (5.0×). This signals that applied AI integration across the 72 domains is where the curriculum produces its highest practical resonance output short of full practitioner certification.

TETRA Curriculum Matrix (RT_Media delivery · STORM_PARTY-(AI & Future Technologies) at 301):

PILLAR / CODE	101 — FOUNDATIONS1.5× Meritcoin	201 — MEASUREMENT 2.0× Meritcoin	301 — APPLIED INTEGRATIONSTORM_PARTY-(AI & Future Technologies)3.0× Meritcoin	401 — ADVANCED PRACTITIONER 5.0× Meritcoin	CERTIFICATION OUTPUT
HEALTH SSHP	Human bioenergetics fundamentals, BERA/ARI baseline measurement, body-as-autonomous-platform (digital-twin framing), THOW health sovereignty module	BERA instrumentation: ARI/ERI clinical calibration, PlayNAC health-twin deployment, IDIPITIS chain measurement, biometric signal processing	AI diagnostics & clinical decision support, wearables ML, population-health modeling, HFVN accessibility integration, BEE nutritional-loop AI, RT_Media health-channel delivery	SSHP system architecture, SCALULAR health-SLA design, REAL_HELP access-number integration, 1000-Year Future Map health layer, COE health-stream qualification	***SSHP Practitioner — qualified to design, operate, and validate HEALTH pillar deployments within a Solid-State Smart-City node***
LAW SSLA	ERES legal architecture fundamentals, Trifurcated Borders (BEST/SOUND/GOOD), NPR	ERES-LEGAL three-instrument stack, MIEVM-LLM as governance commentary, BERA/RCI as governance	AI in legal systems: contract automation, regulatory intelligence, admissibility frameworks, ERES-ADMISSIBILIT	SSLA system architecture, constitutional gift prior-art deployment, Hague filing strategy,	***SSLA Practitioner — qualified to design, operate, and validate LAW pillar deployments and***

	(Non-Punitive Remediation) principles, CBGMODD seat roles overview	tuning, Petition for Rulemaking pathway (16 CFR §1.31 / 12 CFR §1075)	Y v3.0 (48-term glossary), AI bias auditing, RT_Media legal-channel delivery	CBGMODD full governance table, Trifurcated Borders implementation, COE law-stream qualification	***apply ERES legal architecture in institutional contexts***
PROTECTION SSPS	Emergency management fundamentals, SCALULAR SSPS architecture, Storm Party readiness baseline, EMCI REEPER principles, neighbor-resilience ($ELF Reliance)	BERA/ERI threat-resonance measurement, MIEVM for safety-system validation, PoR-F (Fidelity) as safety guarantor, EarnedPath Storm-claim credentialing	AI in security & emergency: threat-intelligence AI, predictive-policing ethics, FDRV autonomous-mobility safety, GSSG infrastructure resilience, Storm Party merit-recognition protocol, RT_Media emergency-broadcast delivery	SSPS system architecture, SCALULAR SSPS full deployment, Storm Party Determination procedure, Council-level emergency governance, COE protection-stream qualification	***SSPS Practitioner — qualified to design, operate, and validate PROTECTION pillar deployments and activate Storm Party protocols at COE level***
SKILLS / TRADE SSST	SCALULAR SSST fundamentals, EarnedPath (EP=CPM×WBS+PERT) credentialing basics, Meritcoin Proof-of-Resonance introduction, 72 Key Domains as vocation map	BERA/RCI workforce resonance, PlayNAC skill-twin deployment, GraceChain credential recording, UBIMIA self-funding curriculum model (students earn while learning)	AI in trades & specialization: domain-specific AI tools across CyberRAVE 72, HFVN trade-access layer, Talonics cybernetic-language substrate, BEE circular-economy trade models, RT_Media skills-channel delivery	SSST system architecture, 48-module 1920-hour curriculum mastery, MIT/SPRU/BAIDU palindromic bridge navigation, COE full qualification, CERT:TETRA-FULL credential issuance	***SSST Practitioner / CERT:TETRA-FULL — the 401 graduate IS a Center of Excellence in human form: qualified to establish, operate, validate, and replicate a COE***

Table note: Purple column (301) = STORM_PARTY-(AI & Future Technologies) programming strand. Green column = Certification output. All 301-level modules delivered via RT_Media channel. Meritcoin yield multipliers: 101=1.5×, 201=2.0×, 301=3.0×, 401=5.0×.

4. Common-Core BEE Stack — Bio-Ecologic Economy & Operational Instruments

The Common-Core BEE stack is the shared economic and operational substrate across all four TETRA pillars. BEE (Bio-Ecologic Economy) is the economic language; THOW, HFVN, FDRV, and GSSG are the operational proof-of-concept instruments that ground BEE in physical reality. Every AI deployment across the CyberRAVE 72 domains is evaluated against BEE alignment at TETRA 301 level.

ACRONYM	EXPANSION	ROLE IN COMMON CORE / AI CURRICULUM	ERES INSTRUMENT LINK
BEE	**Bio-Ecologic Economy**	The economic substrate of the Common Core — value grounded in biological and ecological health rather than extraction. BEE is the economic language that all four TETRA pillars share. AI systems trained within the BEE framework optimize for bio-ecological resonance, not GDP-proxy metrics.	*BERA indices (ARI, ERI, RHC, RCI) as BEE measurement instruments; Meritcoin as BEE currency; REAL=(E·M·R)/(T·S) as BEE formula spine*
THOW	**Tiny House on Wheels**	The VLSA proof-of-concept unit: if the complete SCALULAR stack functions in a THOW (smallest viable deployment), it	*VLSA Principle: 'If it works in a THOW, it works on a generation ship.' Every AI domain application is*

		functions on a generation ship (largest). THOW is both a literal housing-sovereignty instrument and the epistemological test-bench for every ERES AI deployment claim.	*THOW-testable before civilization-scale deployment.*
HFVN	**Hands-Free Voice Navigation**	The accessibility and mobility layer of the Common Core. HFVN provides spoken-traversal of the Talonics cybernetic-language substrate, making the full ERES AI architecture operable by persons regardless of physical mobility, literacy, or device access. HFVN is the equity gateway.	*Talonics + HFVN = New Global Cybernetic Language substrate. AI voice interfaces across all 72 domains must meet HFVN accessibility standard as TETRA 301 deployment requirement.*
FDRV	**Fly & Drive RV**	The advanced mobility proof-of-concept: autonomous, multi-modal personal transport that is simultaneously a home, a mobile COE node, and a storm-response vehicle. FDRV demonstrates SCALULAR SSPS + SSHP + SSST convergence in a single platform.	*BAIDU Apollo (autonomous systems) → FDRV at urban scale. MIEVM validates FDRV safety systems. PoR-F (Fidelity) as FDRV safety guarantor. TETRA 401 graduates are qualified FDRV deployment architects.*
GSSG	**Green Solar-Sand Glass**	The infrastructure-resilience substrate of the Common Core. GSSG converts abundant raw materials (sand, solar) into structural glass and energy simultaneously, providing the physical layer for Storm-Party-resilient smart-city infrastructure that is self-regenerating and zero-extraction-dependent.	*SPRU governance design for GSSG policy environment. REAL formula: GSSG closes the energy-matter loop (E·M). TETRA 301 PROTECTION module specifies GSSG as mandatory storm-resilience infrastructure standard.*

5. CBGMODD — Seven-Seat AI Governance Architecture

CBGMODD defines the seven governance stations through which AI development, deployment, validation, and remediation flow within the ERES framework. All 72 CyberRAVE domain deployments operate through this governance table. Seat 7 is always Diplomat — never a repeated Dignitary. This is a canonical lock (SPT Papers, March 2026).

SEAT	STATION	ROLE IN AI GOVERNANCE	ERES INSTRUMENT
1	**Citizen (C)**	The enrolled SSSC baseline participant. Primary beneficiary of all four SCALULAR pillars.	*UBIMIA enrollment; BERA baseline measurement; EarnedPath credit origin*
2	**Business (B)**	Commercial entities operating within the CyberRAVE 72 domain economy. Domain AI deployment partners.	*Meritcoin value-accrual; MIEVM validation of business AI systems; CCAL licensing*
3	**Government (G)**	Municipal, state, federal, and supranational governance layers. SCALULAR SLA counterparty.	*ERES-LEGAL stack; Constitutional gift prior-art; Petition for Rulemaking pathway*
4	**Media (M)**	RT_Media delivery channel. The TETRA curriculum distribution infrastructure across all 72 domains.	*RHC audience-resonance measurement; MIEVM media-truth validation; STORM_PARTY broadcast layer*
5	**Organization (O)**	NGOs, COEs, research institutions, and civil-society bodies. MIEVM ensemble participants.	*MIEVM node roles; H2C2H synthesis instruments; ERES-ADMISSIBILITY governance*
6	**Dignitary (D)**	Recognized authorities in domain knowledge: academics, elders, specialists. MIEVM validators.	*PoR-A (Authenticity) credential; EAAP authentication; CERT:TETRA-FULL holder qualification*

7	**Diplomat (D)**	The bridge seat. Facilitates cross-domain, cross-cultural, cross-institutional resonance. Seat 7 is always Diplomat — never a repeated Dignitary.	*DAL hinge alignment; BERA/ECR convergence facilitation; MIEVM GO-threshold arbitration*

6. FAVORS — Governance Cycle with O=ODOR (Non-Optional)

FAVORS maps the six primary sector roles in the ERES AI governance cycle. The seventh element — O=ODOR (Output·Delivery·Outcome·Resonance) — is non-optional: it confirms that the governance cycle completes only when resonance is measured and returned to the system. A FAVORS cycle without ODOR is an open loop; open loops are the epistemological disguise that ERES is constitutionally designed to close.

F	EXPANSION	FUNCTION IN AI GOVERNANCE CYCLE
F	**Finance**	Economic resonance layer — Meritcoin/BEE/UBIMIA
A	**Agriculture**	Primary-industry BEE substrate — food, soil, lifecycle loops
V	**Vocation**	Skills/Trade SSST — EarnedPath credentialing, 72 Domain vocations
O	**Organization**	COE / civil-society MIEVM node — governance and validation layer
R	**Research**	MIEVM ensemble participant — scientific validation, H2C2H methodology
S	**Security**	SSPS / Storm Party — protection, resilience, EMCI REEPER
O	**ODOR**	*Output–Delivery–Outcome–Resonance: the non-optional FAVORS O expansion. Confirms that FAVORS completes its cycle only when resonance is measured and returned to the system.*

7. BERA — AI Measurement Indices

The four BERA indices are the measurement instruments through which all AI performance within the ERES framework is evaluated. They feed directly into Meritcoin value accrual (Proof-of-Resonance) and into the MIEVM convergence calculation (ECR). RHC always expands to Resonant Harmony Cycle — never 'Cybernetics' or any other expansion.

INDEX	EXPANSION	FUNCTION IN AI MEASUREMENT
ARI	**Authenticity Resonance Index**	Measures the degree to which an AI system's outputs are authentic to its stated purpose and verifiable against its training and governance documentation. Primary EAAP/PoR-A measurement instrument.
ERI	**Equilibrium Resonance Index**	Measures systemic balance — whether an AI deployment maintains equilibrium across the domains it touches (economic, ecological, social). The BEE economic layer is ERI-grounded.
RHC	**Resonant Harmony Cycle**	The temporal resonance instrument: measures whether AI system behavior returns to harmonic equilibrium after perturbation. Always expands to Resonant Harmony Cycle — never 'Cybernetics' or any other expansion. Finds its most literal expression in the MUSIC domain (45).
RCI	**Register of Collective Indices**	The aggregate resonance registry: the running record of ARI, ERI, and RHC scores across all MIEVM validation events. RCI is the source from which CyberRAVE 72 Key Domains were originally culled. The constraint-based culling epistemology that produced the 72 is the same methodology that governs MIEVM ensemble selection.

8. Domain Cluster Legend

	Commerce & Media		Education & Human Development
	Finance & Insurance		Governance & Public Service
	Health & Life Sciences		Law & Justice
	Primary Industries		Technology & Infrastructure
	Transport & Logistics		Transport & Defense

9. CyberRAVE 72 Key Domains — AI Specialization & ERES Framework Mapping

Source: ERES_Industry-Research.xls, CyberRAVE column, XLS-verified 72 entries. Column header artifact '(70)' in source file is superseded by direct cell count. Canonical count = 72 (gematria value of 'Joseph Allen Sprute': DOB Oct 26 1964 → 10+26=36 → 36×2=72 → 72×3=216 = full name gematria).

#	DOMAIN	AI SPECIALIZATION FOCUS	ERES FRAMEWORK HOOK
01	**ADVERTISING**	Personalization engines, ethical targeting, behavioral ML, ad-fraud detection	*MIEVM resonance scoring; BERA/RCI audience-fit measurement*
02	**AEROSPACE**	Autonomous navigation, predictive maintenance, mission-planning AI, UAV control	*SCALULAR SSPS; PoR-A for safety-critical authentication*
03	**AGRICULTURE**	Precision agriculture, crop-yield prediction, pest detection, soil-analysis ML	*REAL formula resource optimization; VLSA closed-loop sustainability*
04	**AID**	Humanitarian logistics AI, disaster-response coordination, needs-assessment modeling	*SCALULAR AID tier; MIEVM for resource-allocation validation under scarcity*
05	**AIRLINE**	Flight optimization, safety anomaly detection, crew scheduling, demand forecasting	*H2C2H operational intelligence; PoR-F (Fidelity) for safety compliance*
06	**APPAREL**	Trend forecasting, size-recommendation AI, sustainable supply-chain modeling	*BERA/ARI brand resonance; CyberRAVE domain-fit scoring*
07	**AUTO**	Autonomous vehicles, traffic AI, predictive maintenance, V2X communications	*SCALULAR SSST; HPE ASCENT arc for AV maturity staging*
08	**BANK**	Fraud detection, credit scoring, AML compliance AI, algorithmic trading	*Meritcoin/PoR value accrual; BERA as alternative creditworthiness metric*
09	**BIO**	Genomics ML, protein folding, drug-discovery AI, CRISPR outcome modeling	*MIEVM multi-instrument clinical validation; EAAP for bio-agent authentication*
10	**BROKERAGE**	Algorithmic trading, portfolio optimization, sentiment analysis, risk modeling	*Meritcoin Proof-of-Resonance: tuning not mining; BERA/ERI market equilibrium*
11	**CABLE**	Content recommendation, churn prediction, bandwidth-optimization AI	*RHC as content-audience alignment metric*
12	**CAREER**	Job-skill matching AI, career-trajectory modeling, labor-market forecasting	*SCALULAR SSST; BERA/RCI workforce resonance indexing*
13	**CARRIER**	Route optimization, load balancing, last-mile delivery AI, fleet intelligence	*REAL=(E·M·R)/(T·S) as logistics efficiency formula*

14	**CASINO**	Responsible-gambling AI, fraud detection, game-fairness auditing	*MIEVM fairness validation; PoR-A (Authenticity) for game integrity*
15	**CHARTER**	Demand-based routing AI, micro-mobility optimization, on-demand fleet management	*H2C2H real-time synthesis; SCALULAR citizen-mobility SLA*
16	**CHEMICAL**	Reaction-prediction ML, safety-hazard modeling, materials-discovery AI	*MIEVM multi-instrument lab validation; REAL formula process optimization*
17	**COATING**	Quality-inspection vision AI, formulation optimization, defect detection	*PoR-P (Purpose) verification for industrial-process AI*
18	**COLLEGE**	Adaptive learning, admissions AI, dropout prediction, curriculum ML	*SCALULAR SSHP; MIEVM educational-outcome validation*
19	**COMPUTER**	Foundation model development, chip-design AI, compiler optimization, OS intelligence	*H2C2H as AI dev methodology; EAAP Runtime Containment for model deployment*
20	**CONSULTING**	Knowledge-synthesis AI, decision-support systems, strategic forecasting	*MIEVM as consulting validation standard; H2C2H client-synthesis protocol*
21	**COURT**	Case-outcome prediction, legal-document AI, evidence analysis, sentencing modeling	*ERES-ADMISSIBILITY framework; Daubert-load-bearing Deterministic Reference stack*
22	**CRIME**	Predictive-policing ethics AI, forensic ML, pattern recognition, recidivism modeling	*ERES-LEGAL three-instrument stack; PoR for algorithmic accountability*
23	**CRUISE**	Itinerary optimization, passenger-experience AI, safety monitoring, fuel-efficiency ML	*SCALULAR citizen-SLA; BERA resonance for hospitality-quality indexing*
24	**DEALER**	Inventory-optimization AI, dynamic pricing, customer-behavior modeling	*CyberRAVE domain-fit scoring; Meritcoin dealer-reputation accrual*
25	**EDUCATION**	Generative tutoring AI, learning-outcome prediction, accessibility tools, EdTech governance	*SCALULAR SSHP; MIEVM as educational AI validation standard; CCAL licensing*
26	**ENERGY**	Grid optimization AI, renewable forecasting, demand response, smart-meter intelligence	*REAL formula as energy-system spine; VLSA closed-loop sustainability*
27	**ENVIRONMENT**	Climate modeling AI, biodiversity monitoring, pollution detection, ESG scoring ML	*ERES sustainability = abundance floor; BERA/ERI ecological resonance indexing*
28	**FEDERAL**	Policy-simulation AI, regulatory compliance automation, inter-agency coordination	*Constitutional gift thesis; ERES prior-art containment for government AI*
29	**FINANCE**	Risk modeling, RegTech AI, financial-inclusion tools, systemic-risk early warning	*Meritcoin/BERA as alternative financial intelligence; MIEVM fintech validation*
30	**FOOD**	Food-safety AI, supply-chain traceability, nutritional optimization, waste-reduction ML	*VLSA principle; SCALULAR citizen-SLA for food access; closed-loop waste modeling*
31	**FREIGHT**	Multimodal logistics AI, customs intelligence, cargo optimization, supply-chain ML	*REAL=(E·M·R)/(T·S) freight-efficiency formula; SCALULAR SSST*
32	**GAMING**	Procedural generation AI, player-behavior modeling, anti-cheat systems, NPC intelligence	*PoR gaming analogy: resonance not score; CyberRAVE domain-fit for game worlds*
33	**GAS**	Pipeline-integrity AI, leak detection, demand forecasting, emissions optimization	*REAL formula resource loop; SCALULAR SSSC Tier 1 utility services*
34	**HEALTH**	Clinical decision support, diagnostic-imaging AI, population-health modeling, EHR intelligence	*SCALULAR SSHP as constitutional health layer; MIEVM clinical AI validation*

35	**HOME**	Smart-home AI, energy management, predictive maintenance, ambient intelligence	*VLSA: if it works in a THOW it works on a generation ship; SCALULAR citizen node*
36	**INSURANCE**	Actuarial ML, claims-fraud detection, risk personalization, catastrophe modeling	*BERA/ARI risk-resonance scoring; MIEVM actuarial-model validation*
37	**JUSTICE**	Criminal-justice AI ethics, bail-decision auditing, court-efficiency modeling	*ERES-ADMISSIBILITY; PoR-F as fairness guarantor; MIEVM GO threshold*
38	**LAW**	Contract AI, regulatory intelligence, legal-research automation, compliance modeling	*ERES-LEGAL three-instrument stack; Petition for Rulemaking 16 CFR §1.31*
39	**LAWYER**	Legal-strategy AI, document-review automation, precedent mining, client-intake tools	*ERES-ADMISSIBILITY v3.0 (48-term glossary); MIEVM as legal AI evaluation standard*
40	**LIVESTOCK**	Animal-health monitoring AI, breeding optimization, feed-efficiency ML, traceability	*REAL formula as bio-resource loop; MIEVM multi-sensor herd-intelligence validation*
41	**MARINE**	Autonomous vessel navigation, ocean-monitoring AI, port logistics, fishery management ML	*VLSA principle extended to maritime; SCALULAR SSPS maritime security*
42	**MEDIA**	Content-moderation AI, misinformation detection, narrative analytics, generative journalism	*MIEVM as media-truth validation; RHC for audience-resonance measurement*
43	**MEDICAL**	Surgical-robotics AI, drug-interaction modeling, clinical-trial optimization, wearables ML	*SCALULAR SSHP; EAAP PoR for medical-device authentication; MIEVM clinical validation*
44	**MOVIE**	Script-analysis AI, audience prediction, VFX automation, distribution optimization	*BERA/RCI cultural resonance; CyberRAVE domain-fit for content strategy*
45	**MUSIC**	Generative composition AI, rights management, mood-matching, catalog intelligence	*RHC (Resonant Harmony Cycle) as literal domain principle; BERA artist-audience resonance*
46	**OLYMPIC**	Athlete-performance AI, anti-doping analytics, event-logistics optimization, accessibility ML	*MIEVM multi-instrument scoring parallel to multi-event judging; PoR for fair play*
47	**POLITICAL**	Policy-impact modeling, public-sentiment AI, disinformation detection, electoral integrity	*Constitutional gift thesis; ERES prior-art as information-commons governance*
48	**PRISON**	Rehabilitation-outcome prediction, facility-management AI, reentry support tools	*PoR-P (Purpose) as rehabilitation-alignment metric; MIEVM outcome validation*
49	**RACE**	Race-integrity AI, betting-fraud detection, performance analytics, safety monitoring	*MIEVM multi-instrument validation; PoR-A (Authenticity) for competition integrity*
50	**RADIO**	Spectrum-optimization AI, audience analytics, content recommendation, emergency broadcast ML	*RHC as radio-resonance literal domain; SCALULAR SSPS emergency broadcast layer*
51	**RAILROAD**	Predictive-maintenance AI, traffic management, freight optimization, safety monitoring	*REAL formula as rail-efficiency spine; SCALULAR SSSC Tier 1 transport*
52	**RAIL**	Urban-transit AI, passenger-flow optimization, accessibility intelligence, micro-mobility ML	*SCALULAR citizen-mobility SLA; HPE ASCENT arc for transit-system maturity*
53	**REALTY**	Property-valuation AI, market prediction, smart-contract automation, zoning intelligence	*BERA/ERI real-estate resonance; Meritcoin property-value accrual model*
54	**RECYCLE**	Waste-stream classification AI, circular-economy modeling, materials-recovery optimization	*VLSA closed-loop principle; REAL formula waste-elimination as abundance floor*
55	**RELIGION**	Cultural-heritage AI, interfaith-dialogue tools, text analysis, community-service optimization	*DAL hinge (spiritual legitimacy tier); ERES epistemic humility as AI-ethics anchor*

56	**RESEARCH**	Scientific hypothesis-generation AI, literature synthesis, reproducibility modeling, peer-review tools	*MIEVM as research-validation standard; H2C2H as AI-assisted research methodology*
57	**RESORT**	Guest-experience AI, revenue management, sustainability optimization, accessibility tools	*BERA/RCI hospitality resonance; SCALULAR citizen-SLA for leisure access*
58	**RETAIL**	Demand-forecasting AI, computer-vision checkout, inventory optimization, personalization ML	*CyberRAVE domain-fit scoring; Meritcoin reputation accrual for retail trust*
59	**SCHOOL**	Early-intervention AI, special-needs support, teacher-assist systems, safety monitoring	*SCALULAR SSHP; MIEVM educational-outcome validation; CCAL open-access licensing*
60	**SCIENCE**	Foundation models for science, simulation AI, lab automation, cross-discipline synthesis	*MIEVM as scientific validation standard; ERES Triune Math as AI-physics bridge*
61	**SEARCH**	Semantic search AI, knowledge-graph construction, retrieval-augmented generation	*H2C2H as search-synthesis methodology; EAAP for source-authenticity verification*
62	**SECURITY**	Threat-intelligence AI, zero-day detection, network-anomaly modeling, identity verification	*EAAP Runtime Containment; SCALULAR SSPS; PoR for security-agent authentication*
63	**SHIPPING**	Container-optimization AI, port-logistics intelligence, customs automation, route ML	*REAL=(E·M·R)/(T·S) shipping-efficiency formula; SCALULAR SSST*
64	**SPORTS**	Performance-analytics AI, injury prediction, fan-engagement tools, officiating support ML	*MIEVM multi-instrument scoring parallel; PoR-R (Resonance) for athlete-peak modeling*
65	**STEEL**	Process-optimization AI, quality-defect detection, energy-efficiency modeling, alloy discovery	*REAL formula as materials-process spine; MIEVM multi-sensor industrial validation*
66	**STORAGE**	Intelligent data-tiering AI, retrieval optimization, cold-storage intelligence, backup ML	*EAAP for data-asset authentication; BERA/RCI storage-system resonance indexing*
67	**TAX**	Compliance-automation AI, audit-risk modeling, tax-planning intelligence, fraud detection	*ERES-LEGAL stack; PoR-F (Fidelity) as tax-reporting accuracy guarantor*
68	**TEACHER**	AI teaching assistants, professional-development tools, classroom analytics, curriculum AI	*SCALULAR SSHP; MIEVM pedagogical-outcome validation; H2C2H as teaching methodology*
69	**TECHNOLOGY**	AI infrastructure, model governance, safety research, alignment tools, compute optimization	*EAAP full stack; MIEVM as AI governance standard; ERES constitutional gift as prior art*
70	**TOURISM**	Destination-recommendation AI, sustainable-travel modeling, cultural-preservation tools	*BERA/RCI destination resonance; VLSA principle for sustainable tourism loops*
71	**VACATION**	Experience-personalization AI, accessibility tools, pricing optimization, safety monitoring	*SCALULAR citizen-SLA for leisure rights; RHC vacation-satisfaction resonance*
72	**WASTE**	Waste-stream AI, landfill optimization, circular-economy intelligence, hazmat monitoring	*VLSA closed-loop as foundational principle; REAL formula waste→abundance conversion*

10. Canonical Notes & Locked Terms

RHC = Resonant Harmony Cycle — never 'Cybernetics' or any other expansion. Locked.

CBGMODD Seat 7 = Diplomat — never a repeated Dignitary. Locked.

FAVORS O = ODOR (Output·Delivery·Outcome·Resonance) — non-optional. Locked.

CyberRAVE = Cybernetic Ratings Abolishing Veiled Exchanges. Locked backronym, SPT Papers, March 2026.

GunnySack = Guaranteed User-group Network Nullifying Yieldless Service: Accredited Certified Knowledge. Locked.

SaleBuilders = Smart-city Adaptive Living Engineering Banishing Untested Infrastructure—Laboratories Delivering Enduring Resilient Systems. Locked.

RAIL (52) and RAILROAD (51) are intentionally distinct. Granularity is meaningful differentiation.

MUSIC (45) is the literal domain of RHC — the resonance principle finds its most transparent expression in musical harmony.

RELIGION (55) is grounded in the DAL hinge: Dalia/Lucy (personal ground, Co-Founder) + HHDL (spacial-tier spiritual legitimacy).

TECHNOLOGY (69) is the meta-domain governing all AI infrastructure and the domain most directly subject to the EAAP full stack.

ERES Triune Math: (1) $C=R\times P/M$ (2) $M\times E+C=R$ (3) $REAL=(E\cdot M\cdot R)/(T\cdot S)$

HPE DESCENT: Reactive→Flat→Linear→Veiled→Yieldless→Untested

HPE ASCENT: Validation→Coordination→Transparency→Circularity→Dimensionality→Sentience

VLSA Principle: 'If it works in a THOW, it works on a generation ship.'

Meritcoin Proof-of-Resonance: 'It's not mining — it's tuning.'

Sustainability in ERES = not a survival constraint but the floor enabling a blank-check for human happiness, health, and safety.